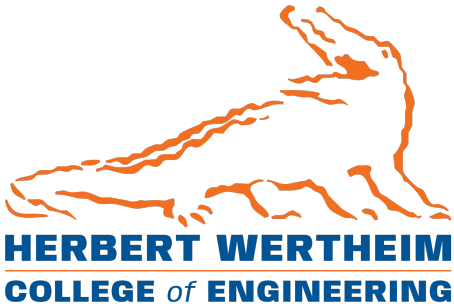
T1: Alpha Test Plan

EEL4908 CpE Design 2

Hardened RISC-V Exam Computer

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1 Scope

This plan is for the alpha build. The pass criterion is **Safe Exam Browser running on the CI-built Debian image** on the Orange Pi RV.

In scope. Image flash and boot; network; build and launch of `seb-linux` on that Debian; basic lockdown once SEB is up.

Out of scope. Instructor control plane (start/stop jobs, fleet UI), Canvas exam completion, and kernel hardening.

Lab baseline. Default login `orangepi`/`orangepi`. UART 115200.

2 Expected behavior

2.1 Board and image

CI in `reptilian-riscv` publishes a Debian `os.img.xz`. `flash-os.sh` writes it to SD. After power, the board reaches a Linux login (UART and SSH). Wi-Fi or Ethernet yields IPv4 and outbound connectivity.

2.2 SEB on Debian

- `seb-linux` builds on riscv64 using the WebKitGTK fallback (no Qt6 WebEngine package).
- SEB launches on the Debian graphical session.
- While running: fullscreen; ordinary desktop exit paths do not dismiss the kiosk; policy-blacklisted processes are refused or terminated.

2.3 `seb-agent` heartbeat

`install-software.sh` copies `seb-agent` to the board. The agent POSTs a JSON heartbeat to a control URL every 5 s.

3 Test procedures

Each test: preconditions, steps, pass/fail.

3.1 Bring-up

ID	Procedure
HW-01	**Flash and boot.** `diskutil list` (never `disk0`); `./scripts/flash-os.sh -device /dev/rdiskN`; power with UART; `screen /dev/cu.usbserial-* 115200`. **Pass:** Linux login, not an initramfs emergency shell.
HW-02	**SSH.** Same LAN; `ssh orangepi@<board-ip>`. **Pass:** session opens; `uname -m` is `riscv64`.
HW-03	**Outbound network.** `ping -c 4` to a known host (or campus DNS). **Pass:** replies, or a documented firewall exception.

Table 1: Bring-up tests.

3.2 SEB

ID	Procedure
SW-01	**Build `seb-linux` on the board.** Pre: HW-02; Qt6 base + WebKitGTK deps. Steps: attempt to find a SEB binary **Pass:** a SEB binary exists. If BUG-01/02 recur, log them and fail this test.
INT-01	**SEB launches.** Pre: SW-01; graphical session. Steps: attempt to launch SEB via terminal commands **Pass:** SEB window/kiosk is visible on the board display.
INT-02	**Lockdown smoke.** Pre: INT-01. Steps: try standard exit shortcuts; optionally start a blacklisted app if policy defines one. **Pass:** cannot leave by those paths; refuse-to-start or terminate matches policy.

Table 2: SEB tests.

3.3 `seb-agent` heartbeat

ID	Procedure
SW-02	**`seb-agent` heartbeat.** Pre: HW-03; laptop HTTP server on port 8000; `install-software.sh` with that `-control-url`. Run `~/seb-agent` on the board. **Pass:** POST `/heartbeat` from the board about every 5 s.

Table 3: `seb-agent` heartbeat.

4 Traceability

Expected behavior	Tests
Debian flash/boot/network (§2.1)	HW-01, HW-02, HW-03
SEB builds, launches, lockdown (§2.2)	SW-01, INT-01, INT-02
`seb-agent` heartbeat (§2.3)	SW-02

Table 4: Expected behavior to tests.